

EXPLORATORY DATA ANALYSIS (EDA) REPORT

Multi-Class Match Result Prediction Using Multinomial Logistic Regression

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Purpose: To understand the structure, distributions, and important pre-match features of the football dataset before training the classification model.

1. Objective of EDA

Exploratory Data Analysis (EDA) is performed before model training to understand the dataset, verify data quality, inspect the target-class distribution, and study the behavior of the main numerical features. For this project, EDA focuses on historical pre-match information used to predict one of three outcomes: Home Win (H), Draw (D), or Away Win (A).

2. Dataset Overview

Property	Value
Dataset file	football_match_prediction_ready.csv
Number of records	20,000
Number of columns	15
Target variable	Result
Target classes	H = Home Win, D = Draw, A = Away Win
Missing values	0 across all 15 columns

The dataset contains match context fields (Date, Country, League, Season, HomeTeam and AwayTeam), engineered historical performance features, and the Result target. The project-ready dataset has no missing values.

3. Main Features Used for Analysis

Feature	Meaning
Home_Recent_Points / Away_Recent_Points	Recent historical points for the two teams.
Home_Avg_Goals_Scored / Away_Avg_Goals_Scored	Historical average goals scored.
Home_Avg_Goals_Conceded / Away_Avg_Goals_Conceded	Historical average goals conceded.
Home_Win_Rate / Away_Win_Rate	Historical win-rate features.
Country / League	Competition context used as categorical information.

4. Target Variable Analysis

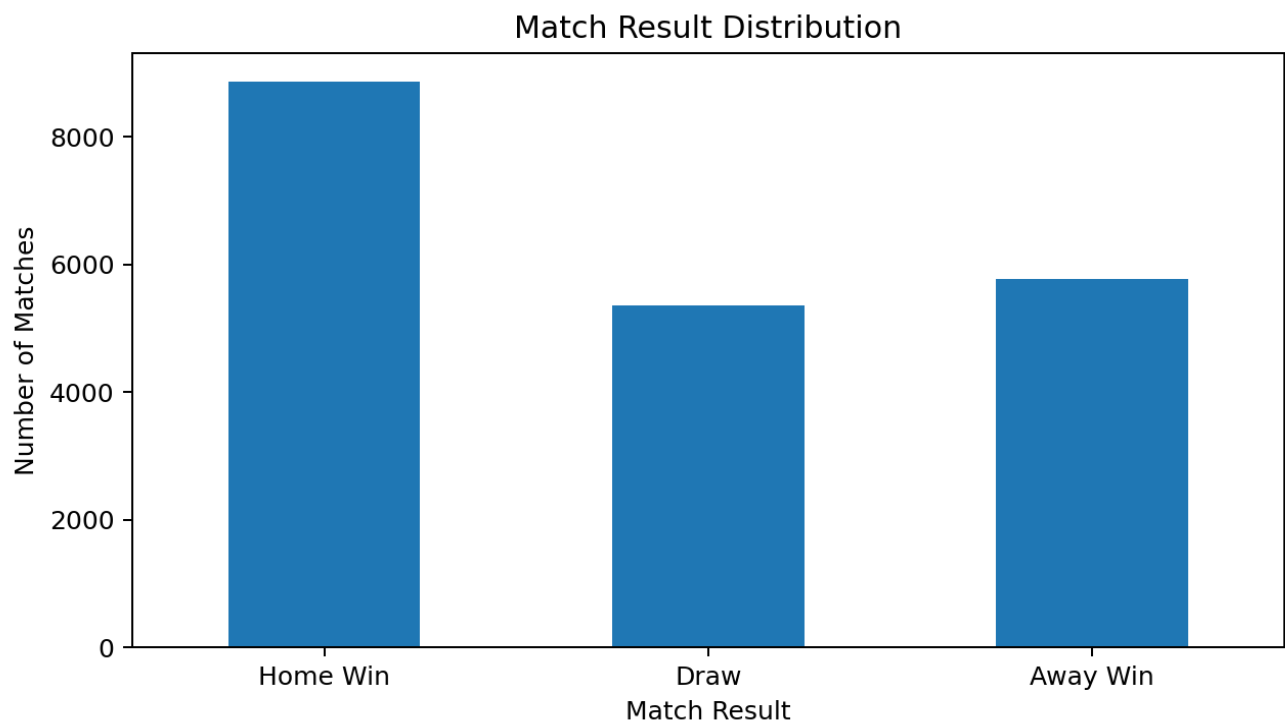


Figure 1. Distribution of the three match-result classes.

Class	Count	Percentage
Home Win	8,867	44.34%
Away Win	5,781	28.90%
Draw	5,352	26.76%

Observation: Home Win is the largest class with 8,867 matches (44.34%). Away Win has 5,781 matches (28.91%), while Draw has 5,352 matches (26.76%). Therefore, the target is not perfectly balanced; Home Win appears more frequently than the other two classes. This is important when interpreting model accuracy and class-level performance.

5. Recent Points Analysis

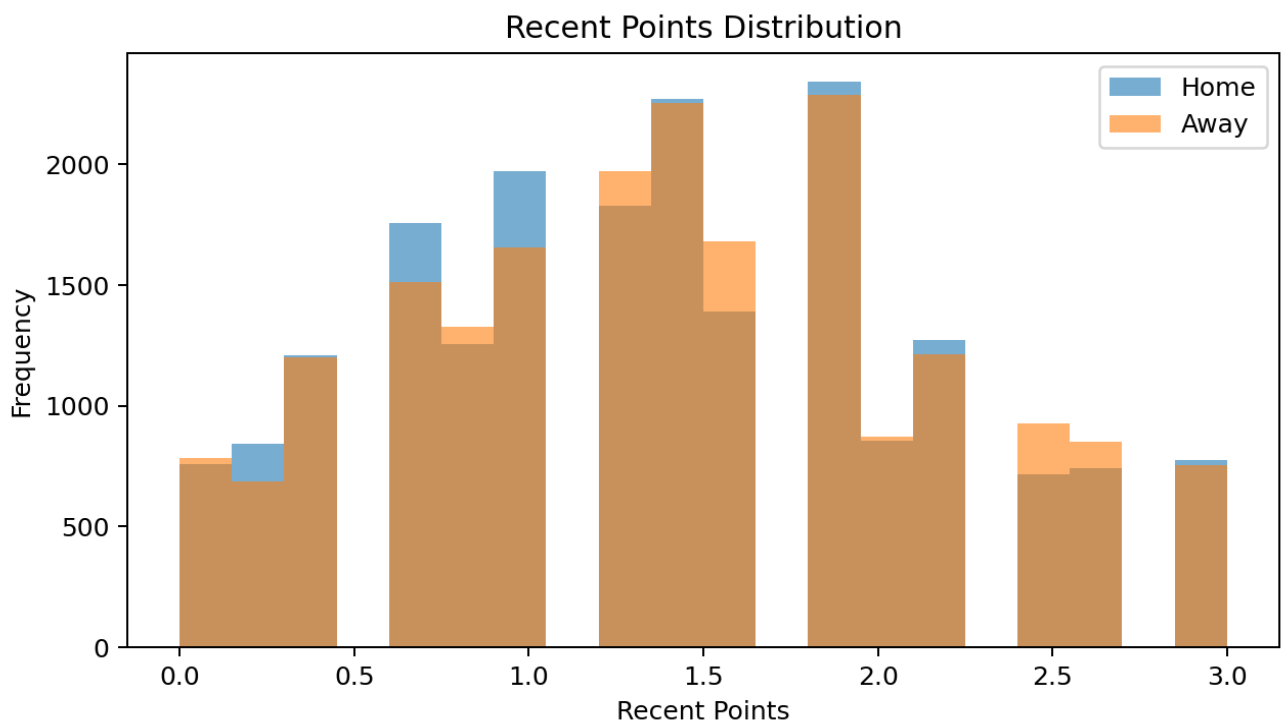


Figure 2. Distribution of Home and Away recent-points features.

Statistic	Home Recent Points	Away Recent Points
Mean	1.350	1.386
Median	1.400	1.400
Minimum	0.000	0.000
Maximum	3.000	3.000

Observation: The recent-points variables summarize recent team form before the match. Comparing the home and away distributions helps identify whether the two sides show similar or different historical form patterns across the dataset. These variables are later supplied to the classification model as numerical predictors.

6. Historical Goal Feature Analysis

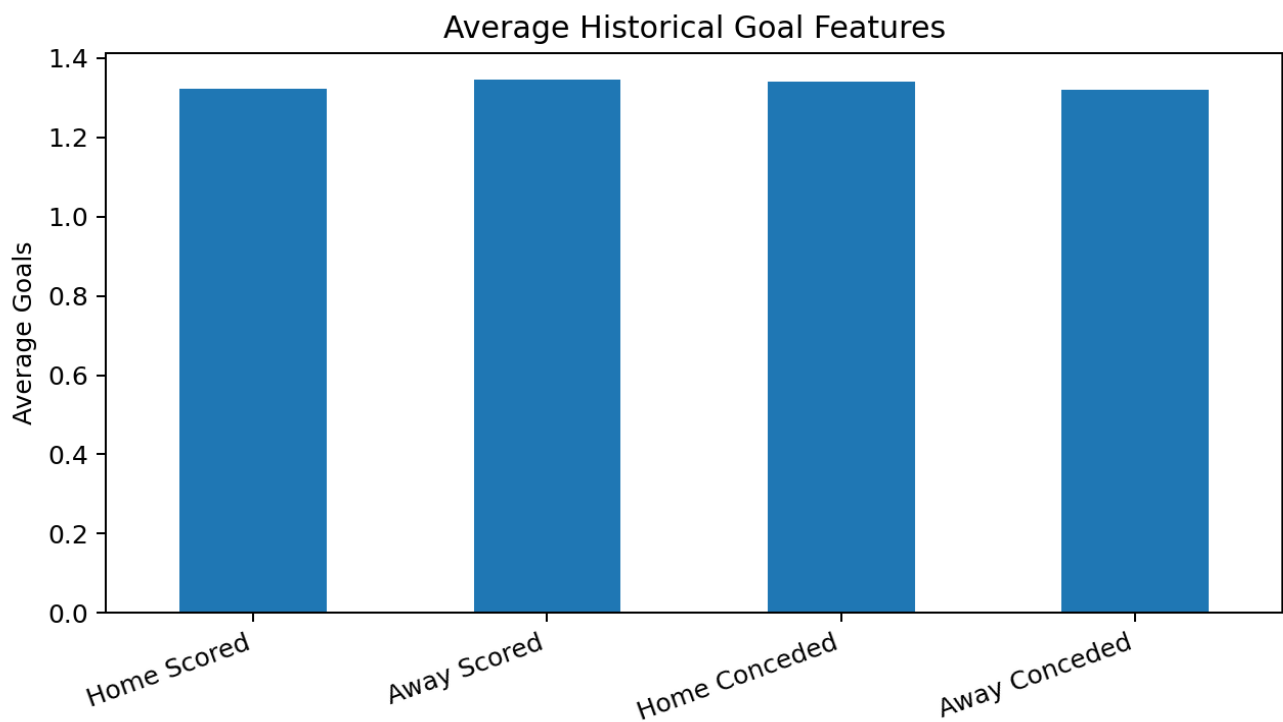


Figure 3. Mean values of historical goals scored and conceded features.

Feature	Mean	Std. Dev.
Home_Avg_Goals_Scored	1.322	0.687
Away_Avg_Goals_Scored	1.345	0.699
Home_Avg_Goals_Conceded	1.342	0.687
Away_Avg_Goals_Conceded	1.320	0.674

Observation: Average goals scored provides information about historical attacking performance, while average goals conceded represents historical defensive performance. The project uses these historical averages rather than the final goals of the match being predicted, avoiding direct leakage of the target result.

7. Historical Win Rate Analysis

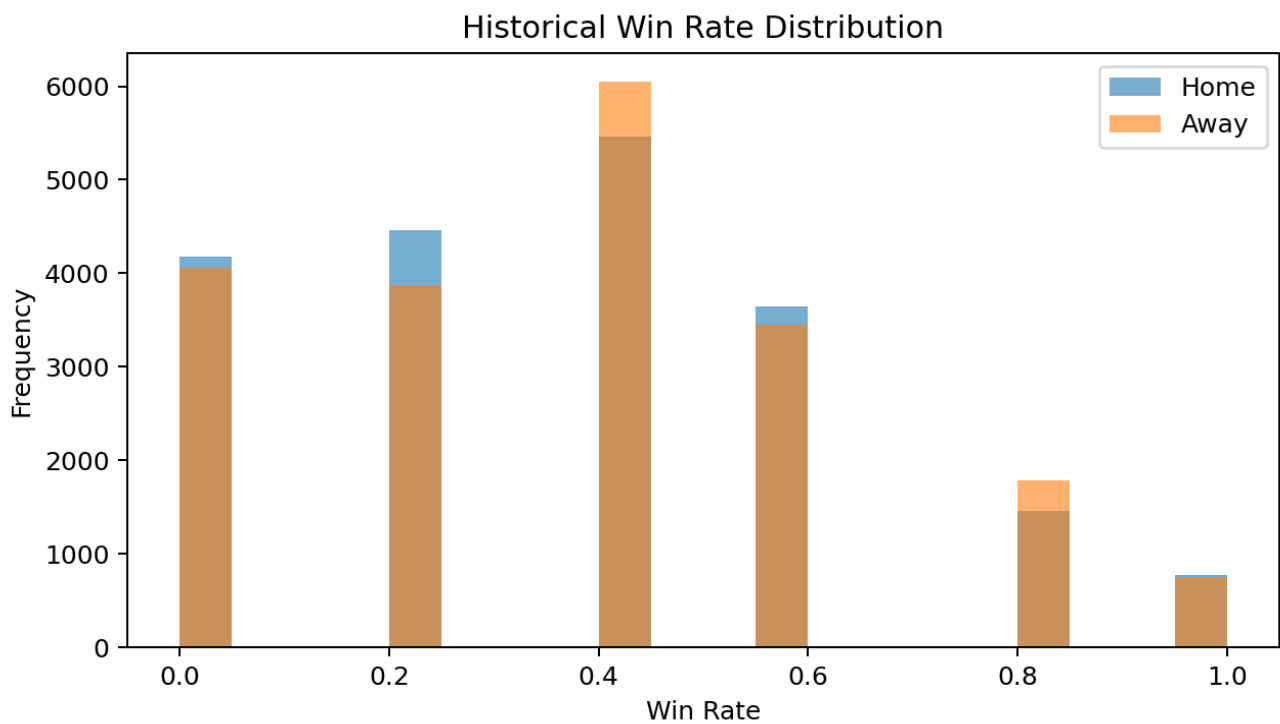


Figure 4. Distribution of Home and Away historical win-rate features.

Statistic	Home Win Rate	Away Win Rate
Mean	0.361	0.373
Median	0.400	0.400
Minimum	0.000	0.000
Maximum	1.000	1.000

Observation: Historical win rate gives a compact measure of previous team success. The home and away win-rate distributions provide useful pre-match information for the classifier and complement recent-points and goal-based features.

8. Key EDA Findings

1. The project-ready dataset contains 20,000 records and 15 columns, with no missing values.
2. The target has three classes: Home Win, Draw, and Away Win.
3. Home Win is the largest class (44.34%), so the class distribution is moderately imbalanced.
4. Recent points, historical goal averages, and historical win rates provide pre-match indicators of team performance.
5. Current-match final goals are not used as predictors because they directly determine the target and would cause data leakage.
6. The EDA supports proceeding to preprocessing and Multinomial Logistic Regression for three-class match-result prediction.

9. Conclusion

The EDA confirms that the dataset is clean and suitable for the planned machine-learning workflow. It also reveals an unequal target distribution, with Home Wins occurring more often than Draws and Away Wins. The selected historical features describe recent form, attacking and defensive performance, and prior win tendency. These observations provide the basis for preprocessing, model training, and evaluation in the next stage of the project.

Next Stage: Data preprocessing -> 80/20 train-test split -> Multinomial Logistic Regression -> Accuracy, Precision, Recall, F1 Score and Confusion Matrix.